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# Sensor-based vehicle count to light usage prediction (simple implementation)

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt


from sklearn.model_selection import train_test_split

from sklearn.pipeline import Pipeline

from sklearn.preprocessing import PolynomialFeatures

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score


np.random.seed(42)


# Simulated sensor data (vehicle count per interval)

n_samples = 240

vehicle_count = np.random.poisson(lam=18, size=n_samples)


# Nonlinear relation: higher vehicle count generally needs more light usage

# Light usage unit can represent % brightness or watt-equivalent index

noise = np.random.normal(0, 2.0, size=n_samples)

light_usage = 0.06 * (vehicle_count ** 2) + 1.8 * vehicle_count + 8 + noise


data = pd.DataFrame({

    "vehicle_count": vehicle_count,

    "light_usage": light_usage

})

data.head()
```

	vehicle_count	light_usage
0	21	73.488333
1	13	43.055015
2	19	62.798998
3	23	79.988364
4	13	40.989897

Next steps: Generate code with data New interactive sheet

```
# Train/test split

X = data[["vehicle_count"]]

y = data["light_usage"]


X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42
```

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)

# Polynomial regression model
poly_degree = 2
model = Pipeline([
    ("poly", PolynomialFeatures(degree=poly_degree, include_bias=False)),
    ("reg", LinearRegression())
])

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

# Performance metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

metrics_df = pd.DataFrame({
    "Metric": ["MAE", "RMSE", "R2"],
    "Value": [mae, rmse, r2]
})
metrics_df
```
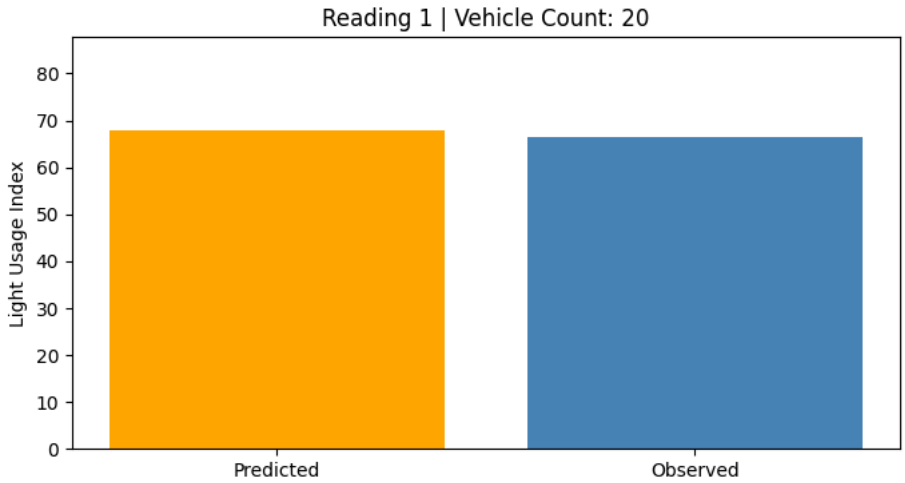

	Metric	Value
0	MAE	1.613077
1	RMSE	2.026286
2	R2	0.988325

Next steps: Generate code with metrics_df New interactive sheet

```
# Real-time style bar chart with anomaly alerts
plt.ion()
fig, ax = plt.subplots(figsize=(8, 4))

# Alert threshold based on training residual standard deviation
train_pred = model.predict(X_train)
train_residual_std = np.std(y_train - train_pred)
anomaly_threshold = 2.5 * train_residual_std

# Simulated incoming sensor stream (latest 30 readings)
stream_counts = np.random.poisson(lam=18, size=30)
alerts = []
```

```
for i, count in enumerate(stream_counts, start=1):
    input_row = pd.DataFrame({"vehicle_count": [count]})
    predicted_light = float(model.predict(input_row)[0])

    # Simulate observed value from physical light sensor
    observed_light = predicted_light + np.random.normal(0, 2.0)

    # Inject occasional anomalies for demonstration
    if i % 10 == 0:
        observed_light += np.random.choice([12, -12])

    deviation = abs(observed_light - predicted_light)
    is_anomaly = deviation > anomaly_threshold
    color = "red" if is_anomaly else "steelblue"

    ax.clear()
    ax.bar(["Predicted", "Observed"], [predicted_light, observed_light], color=["orange", color])
    ax.set_ylim(0, max(predicted_light, observed_light) + 20)
    ax.set_title(f"Reading {i} | Vehicle Count: {count}")
    ax.set_ylabel("Light Usage Index")

    if is_anomaly:
        alert_msg = (
            f"ALERT at reading {i}: vehicle_count={count}, "
            f"pred={predicted_light:.2f}, obs={observed_light:.2f}, deviation={deviation:.2f}"
        )
        print(alert_msg)
        alerts.append(alert_msg)

    plt.pause(0.25)

plt.ioff()
plt.show()

print(f"Total alerts: {len(alerts)}")
```




```
ALERT at reading 10: vehicle_count=13, pred=41.39, obs=52.90, deviation=11.50
ALERT at reading 20: vehicle_count=11, pred=35.08, obs=48.55, deviation=13.47
ALERT at reading 30: vehicle_count=21, pred=72.07, obs=61.07, deviation=11.00
Total alerts: 3
```

```
# Write performance metrics to CSV

from pathlib import Path



metrics_output_path = Path.cwd() / "performance_metrics_week8.csv"

metrics_df.to_csv(metrics_output_path, index=False)



print("Performance Metrics")

print(metrics_df)

print(f"Saved metrics to: {metrics_output_path}")
```

```
Performance Metrics
  Metric     Value
0    MAE  1.613077
1   RMSE  2.026286
2     R2  0.988325
Saved metrics to: /content/performance_metrics_week8.csv
```